Engineering Economics — Interest & Equivalence

DO THIS FIRST, EVERY TIME

Setup & Symbols

sym	meaning
`P`	present amount (t = 0)
`F`	future amount (t = N)
`i`	rate per period (decimal)
`N`	number of periods
`A`	equal payment / period

```
i = r / m        N = m · (years)
m: yr 1 · semi 2 · qtr 4 · mo 12 · day 365
```

- Percent → decimal: 2.75% → **0.0275**.
- Keep **i and N on the same period**.
- **Don't round i early** — only the final answer.

ONE LUMP SUM

Single Payment

```
F = P (1 + i)^N        -- F given P
P = F / (1 + i)^N      -- P given F
simple interest: F = P (1 + i·N)
```

Solve for the unknown

```
N = ln(F/P) / ln(1 + i)
i = (F/P)^(1/N) − 1
double: N = ln 2 / ln(1 + i)
```

Round period counts **up** when the goal is "at least" reach a target.

COMPOUNDING FREQUENCY

Nominal vs Effective

```
i_eff = (1 + r/m)^m − 1
```

- 10% comp. monthly → **10.47%** effective.
- More frequent compounding ⇒ higher effective rate.
- Use i_eff when the **payment period** (e.g. yearly) differs from the **compounding period** (monthly).

Two classic mistakes

- **2.75 instead of 0.0275** → billion-dollar answer.
- **Annual i with N in months** → use `i = r/12`.

EQUAL PAYMENT A EACH PERIOD

Uniform Series (Annuity)

A at the **end** of each period:

```
F/A  F = A·[((1+i)^N − 1) / i]
A/F  A = F·[ i / ((1+i)^N − 1)]
P/A  P = A·[((1+i)^N − 1)/(i(1+i)^N)]
A/P  A = P·[ i(1+i)^N/((1+i)^N − 1)]
```

- Pick the factor whose **two known letters** match.
- **Annuity due** (pay at START): `× (1 + i)`.
- Yearly A, monthly compounding → use **i_eff** with N in years.

BORROW & REPAY

Loans & Remaining Balance

```
payment:  A = P·[i(1+i)^N / ((1+i)^N − 1)]
total paid = A · N
interest   = total − P
```

Payoff partway through

```
balance = P/A on payments STILL OWED
 n = (total payments) − (payments made)
```

Trap: use **payments left (n)**, not payments already made.

PICK THE RIGHT TOOL

Factor Chooser

have → want	factor
P → F	`F/P` compound amt
F → P	`P/F` present worth
A → F	`F/A` series compound
F → A	`A/F` sinking fund
A → P	`P/A` series present
P → A	`A/P` capital recovery

Always draw the **cash-flow timeline**; mark start vs end of period before choosing.

Worked Examples — One of Each Type

numbers you can check against

F/P · COMPOUNDING

Single Deposit

```
$5000, 3%/yr, 5 yr
 annual : 5000(1.03)^5        = $5,796.37
 monthly: 5000(1+.03/12)^60   = $5,808.08

$5000, 20%/yr, 5 yr
 annual : 5000(1.20)^5        = $12,441.60
 monthly: 5000(1+.20/12)^60   = $13,479.85
 simple : 5000(1+.20·5)       = $10,000.00
```

SOLVE FOR P, N, I

Rearranging

```
P now for $5000 in 5 yr @20%/yr:
 P = 5000/1.20^5            = $2,009.39

$2000 → $4000 @20%/yr, how long?
 N = ln2/ln1.20             = 3.80 yr

$2000 → $4000 in 5 yr, what rate?
 i = 2^(1/5) − 1            = 14.87%
```

EFFECTIVE RATE · DON'T ROUND EARLY

Nominal vs Effective

```
CD: $5000, 2.75%/yr nominal,
    comp. monthly, 5 yr
 F = 5000(1+.0275/12)^60   = $5,736.11

Effective of 10%/yr monthly:
 (1+.10/12)^12 − 1         = 10.47%

$10,000 @10%/yr monthly, 50 yr:
 10000(1+.10/12)^600       = $1,453,699.23
```

F/A · ANNUITY

Regular Saving

```
$1000 / yr-end, 10%/yr, 30 yr:
 1000·[(1.10^30 − 1)/.10]  = $164,494.02

$80 / mo-end, 10%/yr monthly, 30 yr:
 i=.10/12, N=360
 80·[((1+i)^360 − 1)/i]    = $180,839.03

Same but pay at START (annuity due):
 180,839.03 · (1+i)        = $182,346.03
```

A/P · PAYMENTS & PAYOFF

Loans

```
$32,000, 6.3%/yr monthly, 4 yr (48):
 A = $755.93/mo   total = $36,284.65

$200,000, 5%/yr monthly, 30 yr (360):
 A = $1,073.64/mo total = $386,511.57

$475/mo, 5.6%/yr monthly, 60 pmts.
Owed after 42nd? n = 18 left:
 475·[((1+i)^18−1)/(i(1+i)^18)]
                           = $8,182.46
```

MULTI-STEP

Combined Cash Flows

```
Double @5%/yr monthly:
 ln2/ln(1+.05/12)=166.7 → 167 mo
                          (13 yr 11 mo)

$80/mo for 10 yr then STOP,
10%/yr monthly, value at 30 yr:
 F10 = 80·[((1+i)^120−1)/i] = $16,387.60
 F30 = 16,387.60·(1+i)^240  = $120,089.53
```

Method: build annuity to year 10, then grow that lump 20 yr (F/P).